

# Mathematics A

## Unit 1 • Chapter OL-T50 Classified

Cross-session • chapter-organised candidate

8 classified items • 35 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

**Dr Eslam Ahmed**

Elite IGCSE Mathematics • [eliteigcse.com](https://eliteigcse.com)

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

# Contents

Unit 1 • Unit 1 • Chapter OL-T50 • 8 items • 35 marks

| Chapter                                 | Items | Marks | Starts |
|-----------------------------------------|-------|-------|--------|
| 01. OL-T50 3D Pythagoras & Trigonometry | 8     | 35    | 4      |

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

# 3D Pythagoras & Trigonometry

Unit 1 • 35 marks • 8 item(s)

June 2025 | 4MA1-2H Q24 | 4 marks

Use midpoint or symmetry structure in a 3D model

January 2022 | 4MA1-1H Q21 | 5 marks

Use midpoint or symmetry structure in a 3D model

January 2020 | 4MA1-1H Q19 | 4 marks

Find angle between a line and a plane

January 2021 | 4MA1-2H Q17 | 3 marks

Find angle between a line and a plane

June 2021 | 4MA1-1H Q22 | 4 marks

Use a non-rectangular base before a line-plane angle

May-Nov 2020 | 2H Q18 | 5 marks

Find angle between a line and a plane

January 2023 | 2H Q23 | 5 marks

Use 3D trigonometry to obtain a terminal volume

November 2025 | 1H Q26 | 5 marks

Use 3D trigonometry to obtain a terminal volume

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

**Working Unit 26**

4MA1-2H Q24

**ORIGINAL QUESTION****4 marks**

24 The diagram shows a square-based pyramid  $ABCDE$

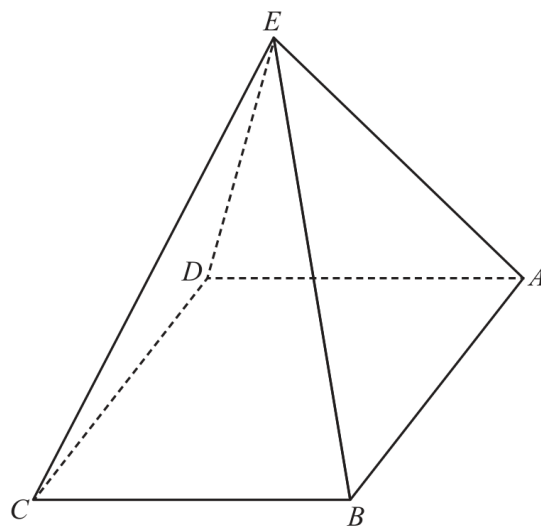


Diagram **NOT**  
accurately drawn

$$EA = EB = EC = ED$$

$M$  is the centre of the horizontal square base  $ABCD$

$Q$  is the midpoint of  $AB$

$$\text{Angle } EQM = 80^\circ$$

$$EA : AB = n : 1$$

Find the value of  $n$

Give your answer correct to 3 significant figures.

$$n = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

**Worked Solution 26**

Three-dimensional trigonometry in a pyramid

**UNIT 1**  
**4 marks****Choose a scale**

$$\text{Let } AB = 1. \text{ Then } QM = \frac{1}{2} \text{ and } AM = \sqrt{(1/2)^2 + (1/2)^2} = \frac{1}{\sqrt{2}}.$$

**Vertical height**

$$EM = QM \tan 80^\circ = \frac{1}{2} \tan 80^\circ.$$

**Sloping edge**

$$EA = \sqrt{EM^2 + AM^2} = \sqrt{\frac{1}{4} \tan^2 80^\circ + \frac{1}{2}} = 2.922474 \dots$$

**Final answer**

$$n = 2.92 \text{ (3 s.f.)}$$

**Original question 17**

4MA1-1H Q21

**ORIGINAL QUESTION****5 marks**

- 21 The diagram shows the prism  $ABCDEFGHJK$  with horizontal base  $AEFG$

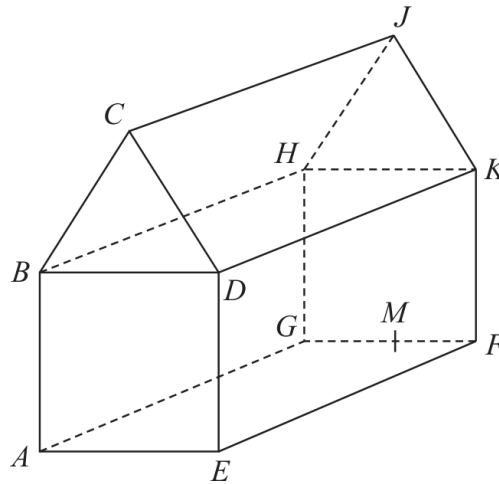


Diagram **NOT**  
accurately drawn

$ABCDE$  is a cross section of the prism where  
 $ABDE$  is a square  
 $BCD$  is an equilateral triangle

$$EF = 2 \times AE$$

$M$  is the midpoint of  $GF$  so that  $JM$  is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of  $T$ , giving your answer in the form  $\frac{\sqrt{p} + \sqrt{q}}{17}$  where  $p$  and  $q$  are integers.

$T = \dots\dots\dots$

**Worked solution 17**

Chapter: 3D Pythagoras &amp; Trigonometry

MODULAR UNIT 1

5 marks

**Q21 – Chapter: 3D Pythagoras & Trigonometry**

Family: Resolve multi-stage solid geometry • Variant: Use midpoint or symmetry structure in a 3D model

**Method**

$$AM = \sqrt{(x^2 + (4x)^2)} = x\sqrt{17}.$$

Why: Use Pythagoras in the horizontal base triangle.

**Method**

$$\text{The equilateral-triangle height is } \sqrt{((2x)^2 - x^2)} = x\sqrt{3}.$$

Why: Use the altitude of the equilateral cross-section.

**Method**

$$\tan y = (x\sqrt{3} + 2x) / (x\sqrt{17}) = (\sqrt{3} + 2) / \sqrt{17}.$$

Why: Use opposite over adjacent for angle MAJ.

**Method**

$$\text{Rationalising gives } (\sqrt{3} + 2)\sqrt{17}/17 = (\sqrt{51} + 2\sqrt{17})/17.$$

Why: Multiply numerator and denominator by  $\sqrt{17}$ .

**Method**

$$\text{Since } 2\sqrt{17} = \sqrt{68}, T = (\sqrt{51} + \sqrt{68})/17.$$

Why: Write the answer in the requested surd form.

**Final answer**

$$T: (\sqrt{51} + \sqrt{68})/17$$

19 The diagram shows a cuboid  $ABCDEFGH$ .

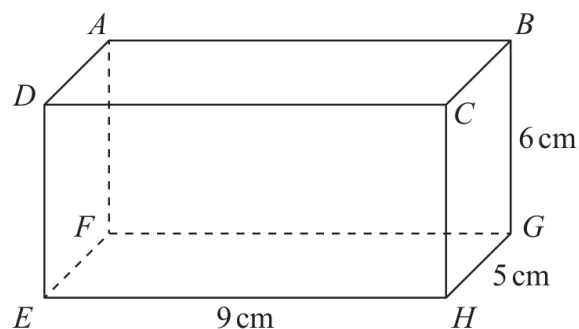


Diagram **NOT**  
accurately drawn

$EH = 9\text{ cm}$ ,  $HG = 5\text{ cm}$  and  $GB = 6\text{ cm}$ .

Work out the size of the angle between  $AH$  and the plane  $EFGH$ .  
Give your answer correct to 3 significant figures.

○



## Worked solution

January 2020 | Question 19 | 4 marks

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

Method / working

1. Horizontal distance AH projection is  $\sqrt{9^2+5^2}=\sqrt{106}$ .
2. Vertical height is 6 cm.
3.  $\tan\theta=6/\sqrt{106}$ .
4.  $\theta=30.2^\circ$  to 3 s.f.

Final answer:  $30.2^\circ$

- (b) Calculate the size of the angle between  $AF$  and the plane  $ADEH$ .  
Give your answer correct to one decimal place.

(3)

## Worked solution

January 2021 | Question 17 | 3 marks

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

Method / working

1. Horizontal distance to the point below F is  $\sqrt{24^2 + 32.5^2} = 40.4$  m.

2.  $AF = \sqrt{40.4^2 + 20^2} = 45.08$  m.

3.  $\sin\theta = 20/45.08$ , giving  $\theta = 26.3^\circ$  to 1 d.p.

Final answer:  $26.3^\circ$

22 The diagram shows a triangular prism  $ABCDEF$  with a horizontal base  $ABEF$ .

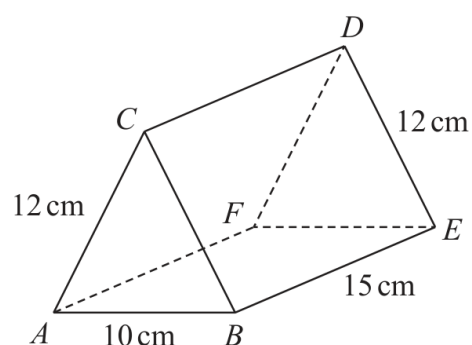


Diagram **NOT**  
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between  $AD$  and the base  $ABEF$ .  
Give your answer correct to 3 significant figures.

..... °

## Worked solution

June 2021 | Question 22 | 4 marks

Family: Resolve multi-stage solid geometry • Variant: Use a non-rectangular base before a line-plane angle

Method / working

1. Use the prism diagram to form the right triangles with lengths 15.8, 19.2 and 10.9 cm.
2. Use  $\tan(\text{angle}) = 10.9/15.8$  (equivalently a sine/cosine relation).
3. Evaluate the inverse trigonometric function.
4. Round to 1 decimal place:  $34.6^\circ$ .

Final answer:  $34.6^\circ$

18 The diagram shows cuboid  $ABCDEFGH$ .

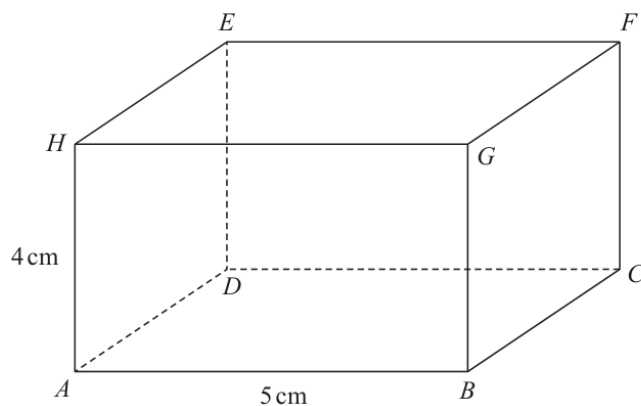


Diagram **NOT**  
accurately drawn

$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

The size of the angle between  $CH$  and the plane  $ABCD$  is  $35^\circ$

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

.....cm<sup>3</sup>

(Total for Question 18 is 5 marks)

## Answer reference | May-Nov 2020 | Question 18

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

|    |                                                                                                                                                                                                                                                                                               |      |   |    |                                                                                       |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|----|---------------------------------------------------------------------------------------|
| 18 | eg $\frac{4}{AC} = \tan 35$ oe $\frac{AC}{4} = \tan 55$ oe $\frac{AC}{\sin 55} = \frac{4}{\sin 35}$ oe <b>or</b><br>$CH = \frac{4}{\sin 35}$ oe (= 6.97...) <b>and</b> $\frac{AC}{6.97} = \cos 35$ oe <b>or</b><br>$CH = \frac{4}{\sin 35}$ oe (=6.97...) <b>and</b> $AC^2 = 6.97^2 - 4^2$ oe |      |   | M1 | A correct trig statement involving AC <b>or</b> trig and then Pythagoras involving AC |
|    | $(AC =) \frac{4}{\tan 35}$ oe eg $(AC =) 4 \tan 55$ (= 5.71...) <b>or</b><br>$(AC =) \frac{4 \sin 55}{\sin 35}$ <b>or</b> "6.97" $\times \cos 35$ oe <b>or</b> $(AC =) \sqrt{6.97^2 - 4^2}$                                                                                                   |      |   | M1 | complete method to find AC                                                            |
|    | $(BC =) \sqrt{5.71^2 - 5^2}$ (= 2.76...)                                                                                                                                                                                                                                                      |      |   | M1 | complete method to find BC                                                            |
|    | $4 \times 5 \times "2.76..."$                                                                                                                                                                                                                                                                 |      |   | M1 | method to find volume                                                                 |
|    |                                                                                                                                                                                                                                                                                               | 55.3 | 5 | A1 | accept 55.1 – 55.5                                                                    |
|    |                                                                                                                                                                                                                                                                                               |      |   |    | <b>Total 5 marks</b>                                                                  |

## Worked solution

May-Nov 2020 | Question 18 | 5 marks

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

Official final mark-scheme pages are embedded immediately after this question.



23 The diagram shows a solid prism  $ABCDEFGHJ$

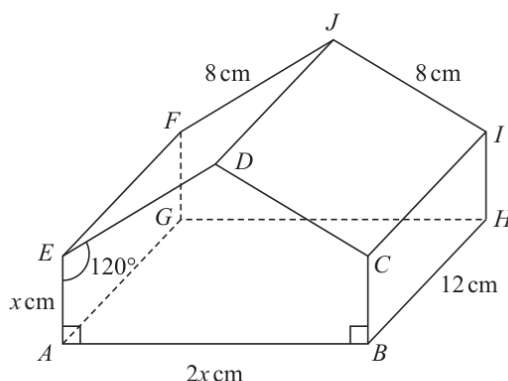


Diagram **NOT** accurately drawn

The prism is such that each cross section is a pentagon where

$$AE = BC = x \text{ cm} \quad AB = 2x \text{ cm} \quad ED = CD = 8 \text{ cm}$$

$$\text{angle } EAB = \text{angle } CBA = 90^\circ \quad \text{angle } AED = \text{angle } BCD = 120^\circ$$

Given that  $AG = BH = EF = DJ = CI = 12 \text{ cm}$

calculate the angle that  $AJ$  makes with the base  $ABHG$  of the prism.

Give your answer correct to 3 significant figures.



(Total for Question 23 is 5 marks)

## Answer reference | January 2023 | Question 23

Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume

| Q  | working                                                                                                                                                                                                                                                                                                                                                    | Answer | Mark | Notes                                                                                                                                                 |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23 | $[DN =]8\sin 30 \text{ or } 8\cos 60 (= 4) \text{ oe [where } N \text{ is the midpoint of } EC]$<br><b>or</b><br>$[x =]8\cos 30 \text{ or } 8\sin 60 (= 4\sqrt{3} = 6.928\dots)$<br><b>or</b><br>$2x = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 120} (= \sqrt{192} = 8\sqrt{3} = 13.85\dots)$                                                     |        | 5    | M1                                                                                                                                                    |
|    | $[DN =]8\sin 30 \text{ or } 8\cos 60 (= 4) \text{ oe eg } \sqrt{8^2 - (4\sqrt{3})^2} (= 4)$<br><b>And 1 of</b><br>$[x =]8\cos 30 \text{ or } 8\sin 60 (= 4\sqrt{3} = 6.928\dots) \text{ oe or}$<br>$\sqrt{8^2 - "4"'^2} (= 4\sqrt{3} = 6.928\dots) \text{ or}$<br>$2x = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 120} (= 8\sqrt{3} = 13.85\dots)$ |        |      | M1 $[(JM =) 4 + 4\sqrt{3} \text{ implies M2}]$                                                                                                        |
|    | $[AM =]\sqrt{12^2 + ("4\sqrt{3}"')^2} (= \sqrt{192} = 8\sqrt{3} = 13.856\dots) \text{ oe}$<br>$[\text{where } M \text{ is the midpoint of } GH]$                                                                                                                                                                                                           |        |      | M1 <b>Clear intention to be AM (not EC)</b>                                                                                                           |
|    | $\tan MAJ = \left( \frac{"4" + "4\sqrt{3}"'}{"8\sqrt{3}"} \right) \text{ oe eg } \tan MAJ = \left( \frac{10.928\dots}{13.856\dots} \right)$<br>if student uses sin or cos, then $AJ = 17.647\dots$ to award marks this must come from a correct method or be correct                                                                                       |        |      | M1                                                                                                                                                    |
|    | <i>Correct answer scores full marks (unless from obvious incorrect working)</i>                                                                                                                                                                                                                                                                            | 38.3   |      | A1 Accept 38.1 – 38.4<br>If no marks scored then award<br>SCB2 for $\tan MAJ = \frac{4+x}{\sqrt{x^2 + 12^2}}$<br>or SCB1 for $AM = \sqrt{x^2 + 12^2}$ |
|    |                                                                                                                                                                                                                                                                                                                                                            |        |      | <b>Total 5 marks</b>                                                                                                                                  |

## Worked solution

January 2023 | Question 23 | 5 marks

Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume

Official final mark-scheme pages are embedded immediately after this question.

26 Here is a prism  $ABCDEFGH$  with a horizontal, rectangular base  $ABGF$

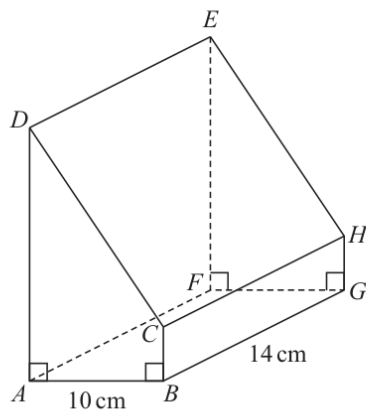


Diagram **NOT**  
accurately drawn

$$AB = 10 \text{ cm} \quad BG = 14 \text{ cm}$$

$$\text{angle } DAB = \text{angle } ABC = \text{angle } EFG = \text{angle } FGH = 90^\circ$$

$$BC = \frac{1}{5} AD$$

The angle of elevation of  $E$  from  $B$  is  $50^\circ$

Calculate the volume of the prism.

Give your answer correct to 3 significant figures.

Show your working clearly.



..... cm<sup>3</sup>

(Total for Question 26 is 5 marks)

## Worked solution

November 2025 | Question 26 | 5 marks

Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume

### Method

The horizontal base ABGF is a rectangle.

$$AB=10, \text{ and } BG=14$$

So the horizontal distance from B to F is

$$BF=\sqrt{10^2+14^2}=\sqrt{296}$$

The angle of elevation of E from B is  $50^\circ$ .

Since E is vertically above F,

$$\tan 50^\circ = \frac{FE}{BF}$$

$$FE=\sqrt{296}\tan 50^\circ$$

The prism has the same cross section throughout, so

$$AD=FE=\sqrt{296}\tan 50^\circ$$

Also

$$BC=\frac{1}{5}AD$$

Area of the trapezium cross section ABCD:

$$\frac{1}{2}(AD+BC) \times AB$$

$$=\frac{1}{2}\left(AD+\frac{1}{5}AD\right) \times 10$$

$$=6AD$$

Volume of the prism:

$$6AD \times 14=84AD$$

$$=84\sqrt{296}\tan 50^\circ$$

$$=1722.311\dots$$

Correct to 3 significant figures:

$$1720$$

Answer:  $\boxed{1720 \text{ cm}^3}$